

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 24

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • B: frost-free period (150 days) exceeds the 120 needed whereas A (110) is too short - late frosts would kill seedlings (1) • B is warmer / longer thermal growing season because south-facing slopes receive more insolation; valley floors are frost pockets where cold dense air collects (1) • But B has lower rainfall and a steeper slope (erosion / machinery difficulties) so irrigation / contour cultivation may be needed (1)	3
01.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Greenhouses / polytunnels heated by solar gain or burning gas / paraffin (ventilated to prevent overheating) (1) • Woven plastic cloches / fleece over rows to retain warm air near the ground (1) • Burning oil / gas or using large fans in orchards to prevent frost damage to blossom (1) • Site selection (aspect, avoiding frost pockets); heated buildings and shelter for livestock (1)	3
01.3	<i>1 mark for each point.</i> • Mammals and birds use food energy to maintain a constant body temperature (thermoregulation) (1) • Reducing heat loss (shelter, heated buildings) means less food energy is used for heat and more is available for growth (1)	2
01.4	<i>1 mark for each point.</i> • Long days increase milk production and egg laying; poultry grow best with short days; sheep mate as days shorten in autumn (1) • Artificial lighting extends the growing season for greenhouse crops or simulates autumn in spring to give sheep a second mating / lambing season (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-309; AQA spec 3.5.1.1.	
02.1	<i>Correct answer only.</i> • Its mass (1)	1
02.2	<i>1 mark for each point.</i> • The 24-hour rotation period means the surface is not exposed to sunlight for long, so temperature extremes / excessive heating and cooling are reduced (1) • The axis is tilted relative to the orbit so the amount of insolation at each latitude changes through the year, producing seasonal variations in conditions (1)	2
02.3	<i>1 mark for each point.</i> • The distance from the Sun gives surface temperatures between 0 and 35 °C in most areas: too close and water would boil / evaporate, too far and it would freeze (1) • Atmospheric pressure (produced by the mass of gases held by gravity) is high enough to stop the water boiling / evaporating away; at much lower pressure there would be no liquid water (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	
03.1	<i>1 mark for each point.</i> • Its reading is adjusted to match a meter known to be accurate so that the results are accurate / comparable (1) • If it cannot be adjusted the percentage error should be recorded so the results can be corrected (1)	2
03.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Droppings: diet, gender, territory size (1) • Owl pellets: diet (1) • Feeding marks (eg chewed hazel nuts): presence of species such as dormice (1) • Nests / burrows: presence and breeding (1) • Tracks / footprints: presence, movements (1) • Territorial marks (scratching posts): presence and territory (1)	3
03.3	<i>1 mark for each point.</i> • Bat detectors / sonograms record the ultrasonic sounds bats produce, which differ between species (1) • The recordings provide evidence of presence, abundance and activity of species that cannot easily be seen (1)	2
03.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Marked individuals mix freely with the rest of the population (1) • Marks do not fall off or wear away, and marking does not affect survival (1) • The population does not change during the study (no births, deaths or migration) (1) • All individuals have an equal chance of being caught / animals are mobile and can be caught (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402, 405, 407; AQA spec 3.1.2.3.5-3.1.2.3.6.	

Question	Marking guidance	Marks
04.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Selective breeding: parents chosen for desirable genetic characteristics (Limousin muscle growth, Merino wool, Cheviot hardiness); combines genes but characteristics of offspring not fully predictable; many generations needed; risk of inbreeding (2) • Asexual reproduction (cuttings, runners, tissue culture, cloning): offspring genetically identical to the parent so characteristics are predictable and survival high; but no variation so no improvement and vulnerability to the same disease; cloning of animals (nucleus transfer into an enucleated egg, surrogate) still developing (2) • Cross-breeding: two breeds combined to give a mix of characteristics with hybrid vigour (heterosis) and lower inbreeding risk, eg Zebu (heat tolerant) x Ayrshire (high milk yield) (2)	6
04.2	<i>1 mark for each point.</i> • AI: semen from a male with desirable characteristics can fertilise many more females than natural mating; frozen semen stored and transported easily and used after the male dies (1) • Embryo transfer: FSH stimulates a superior female to release many eggs, which are fertilised (IVF) with sperm from a superior male (1) • Each embryo is implanted in a different surrogate female (1) • So the donor produces many offspring per year rather than one per pregnancy (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-323; AQA spec 3.5.1.2.	
05.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Light: selective felling to create clearings for plants needing high light while retaining dense canopy for shade species; clearings also give warm sand / soil for lizard eggs; leave dead wood; woodland floor plants (bluebells) grow before the canopy leafs (2) • Dissolved oxygen: narrowing a slow river increases flow and turbulence so more oxygen dissolves, suiting trout and mayfly larvae; shallow water warms quickly for egg development (2) • Nutrients: removing fertile topsoil / not fertilising allows plants of nutrient-poor soils (eg sundew on acidic nitrate-deficient soils) that cannot compete on fertile ground; dung patches add local nutrients (2)	6
05.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Food: wildflower grasslands for seed-eating birds; providing food species (1) • Control of predation: island nest sites unreachable by foxes; trapping / culling / eradicating introduced rats (Scilly Isles - Manx shearwater and storm petrel breeding success increased) (1) • Control of competitors: removing rhododendron so wildflowers get light (1) • Pollination: a range of flowering plants with different flowering times so pollinators have food all season (1) • Seed dispersal species (hippos, elephants, monkeys) conserved; control of introduced pathogens (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 43-46; AQA spec 3.1.2.3.4, 3.1.3.1.	

Question	Marking guidance	Marks
06.1	<i>1 mark for each point.</i> • Small clearings were farmed for a few years then abandoned as nutrients declined; the forest recolonised and soil fertility recovered before the area was cleared again (1) • Population growth shortens the gap between clearances so the forest and soil do not recover (1) • Clearings enlarge and degrade; less regrowth (1)	3
06.2	<i>1 mark for each point.</i> • Rainforest soils are thin and nutrient-poor (nutrients held in the biomass); once cleared, fertility falls in a few years and yields drop, so more land is cleared (1) • Abandoned land may not be recolonised if no forest survives nearby; erosion of exposed soil (1)	2
06.3	<i>1 mark for each point.</i> • Mining (aluminium, iron, gold, copper) destroys habitat directly and pollutes rivers with turbid drainage, cyanide and mercury from processing (1) • Reservoirs (eg Tucurui, 2850 km²) flood large areas, block migratory fish and river dolphins; decomposing vegetation produces acidic, deoxygenated water; fluctuating downstream flows (1) • Droughts increase frequency, area and duration of fires (long fires boil sap and kill trees); cloud forests lose humidity as clouds form higher so frogs and epiphytes decline; some models predict replacement by savannah (1)	3
06.4	<i>1 mark for each point.</i> • A creditor country or conservation charity cancels / buys part of a debtor country's foreign debt (1) • In return the country guarantees to protect an area of rainforest (eg USA - Guatemala, Brazil; WWF - Costa Rica, Philippines) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 55-57; AQA spec 3.1.2.3.4.	
07.1	<i>1 mark for each point.</i> • In winter nutrients are high (storms disturb sediments, deep mixing) but light is too low for photosynthesis (1) • In spring light rises while nutrients are still high so growth is rapid - the spring bloom (peak about May) (1) • The bloom uses up nutrients (falling to 10 by July); dead plankton sink out of the photic zone carrying nutrients to the seabed (1) • So nutrients become limiting in summer despite high light; a smaller autumn bloom follows when storms return nutrients while light is still adequate (1)	4
07.2	<i>1 mark for each point.</i> • Warm surface water is stable (no seasonal mixing / storms) so nutrients lost to deep water in sinking dead organisms are not replaced (1) • Nutrient (especially phosphate) concentrations in the photic zone remain very low - the limiting factor (1)	2

Question	Marking guidance	Marks
07.3	<i>1 mark for each point.</i> • Upwelling (offshore winds, seamounts, currents deflected upwards) brings cold nutrient-rich deep water into the photic zone (1) • Coastal waters receive nutrients washed off the land in rivers and storms disturb shallow sediments (1) • Nutrients plus light in the shallow photic zone allow high algal productivity supporting rich food webs and fisheries (eg Peru anchovy) (1)	3
07.4	<i>1 mark.</i> • Phosphates have low solubility and no atmospheric reservoir; they are only added by river runoff or upwelling and are lost to sediments (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-339; AQA spec 3.5.2.1.	
08.1	<i>1 mark for each point.</i> • Irrigation depletes rivers and aquifers (eg Aral Sea, Ogallala); increases evaporation (1) • Soil erosion and compaction reduce water retention and infiltration so runoff and river flow fluctuations increase (1) • Evapotranspiration changes: increased in irrigated arid areas, decreased where forests were cleared (1)	3
08.2	<i>2 marks for each point (partial credit for a partly correct point).</i> • Seasonal / local: avoids energy for heating, lighting greenhouses out of season and long-distance transport (food miles) and the pollution this causes (2) • Organic: uses natural processes (rotation, legumes, manure, biological control, predator habitats) instead of artificial fertilisers and pesticides, so less pollution, energy use and harm to non-target species; free-range and Fairtrade also accepted for welfare / social reasons (2)	4
08.3	<i>1 mark for each point.</i> • Food shortages after 1945 (dependence on US food aid); farmers could not afford loans to invest so grants funded machinery, drainage, hedgerow removal, liming, improved livestock (1) • Guaranteed market / price support: government bought surpluses to hold prices up (and sold stores in poor years) giving farmers security to increase output (1) • Consequences: loss of hedgerows and wetlands, intensification, pesticide and fertiliser pollution; later food surpluses (butter mountains) that could not be sold without harming LEDC producers (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-333; AQA spec 3.5.1.3-3.5.1.4.	
09.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Image databases: identifying individuals (tiger stripes, whale shark spots, dolphin fins) to find territory size, movements, lifespan, social groups - few species have unique recognisable features (2) • eDNA: detecting the presence of a species from DNA shed into water (great crested newt, invasive carp) - shows presence not abundance; contamination / DNA transport by water (2) • Geolocator tags: tracking small birds on migration where transmitters are too heavy - only record light and time so position is approximate; animal must be recaptured to download data (2)	6

Question	Marking guidance	Marks
09.2	<i>1 mark for each point.</i> • Microwaves emitted by the satellite reflect off the surface giving data on wave height, wind velocity, sea surface altitude and oil slicks (which smooth the surface) (1) • Microwaves pass through cloud so monitoring is not interrupted by weather (1)	2
09.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Droppings: diet, gender, territory size (1) • Owl pellets: diet (1) • Feeding marks (chewed hazel nuts - dormouse), tracks / footprints, nests / burrows, scratching posts (territory) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.7.2.3.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Resources: wood, fibres (cotton, wool, silk), oils, fuels (wood, biofuels), food; new food species (morama bean, perennial cereals, eland, bison) better adapted to local conditions • Biomimetics: vehicle design, architecture, structures, adhesion, materials, ultrasound • Medicines from plants and animals (examples); most species unstudied - argument for habitat conservation • Physiological research using wildlife species; wildlife as pest control agents (ladybirds, Encarsia, Cactoblastis, Bt) • Genetic resources: crop wild relatives, Vavilov centres, gene pools • Ecosystem services: atmospheric composition, hydrological and biogeochemical cycles, soil maintenance, pollination, seed dispersal, habitat provision • Ethical / aesthetic / recreational arguments; the unknown value of species not yet investigated (precautionary principle) • Evaluation: which arguments are most persuasive to decision makers; economic valuation vs intrinsic value; judgement	9
10.2	<i>1 mark.</i> • The variety and abundance of life (species, genes, habitats) in an area / on Earth (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which agriculture can be made sustainable while feeding a growing human population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Need: population possibly 12 billion; humans raise carrying capacity; trophic level of diet • Impacts to be reduced: habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • Social, economic and political influences: consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • Strategies: cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • Limits: little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25

Question	Marking guidance	Marks
11.2	Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Temperate broadleaf woodland: seasonality, deep soils, high biodiversity, resources, climate and soil regulation; deforestation, fragmentation, management change; SSSIs, ancient woodland protection, coppicing, mixed planting • Tropical rainforest: stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils; fuelwood, timber, agriculture, mining, reservoirs, climate change; protected areas, debt-for-nature swaps, sustainable exploitation • Tropical coral reefs: cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity; fisheries, erosion protection, medicines, tourism; damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species; MPAs, no-take zones, control of activities • Deep-water reefs: cold, dark, slow growth; research, fisheries; trawling, oil / gas, acidification; SACs • Oceanic islands: isolation, endemism, no mammal predators; exploitation, introduced species, habitat change, sea level; eradications, control of development and visitors • Mangroves: halophytes, anaerobic mud; erosion protection, fisheries nurseries, sediment trapping; clearance for aquaculture / urban / ports, pollution, climate change; protection, replanting • Antarctica: extreme cold, seasonality, marine nutrients, ice; water store, albedo, carbon, research; climate change, ozone, tourism, fishing, minerals; Antarctic Treaty, CCAMLR, tourism and waste rules • Synthesis: features -> importance -> threats -> strategy; evaluation of success; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	25